Assignment : Your first full game (10%)

Due date : Octobre 18 2026
Individually

This assignment tasks you to make a simple platformer

> **You need to create an assignment repo by going to https://github.com/amcnabbbaltar/420_541_Assignement_2026 and clone the template and create your own fork.**

1. Menu System

- **Main Menu:**
  - **Play Button** → Loads the **first level**.
  - **Quit Button** → Exits the game.
- **End Screen:**
  - Displays the **final score**.
  - **Restart Button** → Resets the score and reloads the **first level**.

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2. Player Mechanics

- **Basic Movement:**
  - The player can **walk, run, jump, and double jump**.
  - You can move in 3 axis (3D movement)
  - **Shift key** toggles **Run**.
    - Your player movement speed doubles for the time the button is pressed down.
  - **Spacebar** triggers **Jump**.
    - The longer you press the space bar, the higher you go.
    - Your player should jump after 3s of holding the spacebar or when the spacebar is released if it is released before the 3 second.
  - **Double Jump** plays a **Flip animation**.
    - Can only trigger in mid air.
    - Double Jump can only trigger while in mid-air and the player has the power up active.

        - The double jump propels you upward at the same speed regardless the amount of time you have pressed the space button for.
        - It will trigger an double jump animation
    - Movement is physics driven (move the object with the Rigid Body)

- **Animations:**
    - Uses **Blend tree** for smooth transitions between **Idle, Walk, and Run**.
    - Double Jump should have a **special animation effect** like a sound or a particle effect when you do the animation.
    - Animation scripting should be on a separate script than player movement.
- **Camera Movement**
    - The intro scene has a camera already set up to follow you. You can simply duplicate the original scene as a base for your subsequent levels.
- **Power-ups:**
    - **Speed Boost Pickup (Orange Pickup)** → Increases movement speed for **5 seconds**.
    - **Score Boost Pickup (Purple Pickup)** → Double the score you get per Score Pickup for 10 seconds.
    - **Jump Boost Pickup (Blue Pickup)** → Enables **Double Jump** for **30 seconds**.
    - **Score Pickup (Yellow Pickup)** → Adds **50 points**.
    - Pickups should:
        - **Hover and rotate**.
        - **Disappear when collected**.
        - Trigger a **Sound** upon collection.
    - When you touch a pickup the pickup should disappear and disable the collision and it should reappear 30 seconds later. For all pickups **except** the score pickup.
    - When the score pickup gets touched it needs to be destroyed.
    - All your pickups needs to be saved as Prefabs.

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3. HUD & UI

- **Score Display:**
    - Score must always be visible in the **top-right corner**.
    - At the end of the game, the **final score** must be displayed.
- **Health System:**

- The player should have **3 health points**.
- **Colliding with a Red Trap removes 1 health point**.
- If the player **loses all health**, the level **restarts**.

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4. Game Structure & Progression

- **Levels & Scene Transitions:**
  - The game should contain **at least 3 levels**.
    - You can duplicate the provided scene to design your levels. In order to keep the camera setup provided.
  - Each level must have a **physical end goal object** (e.g., portal, door, checkered flag).
  - **Interacting with the end goal** should transition the player to the **next level**.
  - The **final goal object** should lead to the **end screen menu**.
- **Traps & Challenges:**
  - **Red Traps** reduce health by **1 point**.
  - Falling off the level **resets the level**
  - Your level design should start easy and then get progressively harder.
    - More traps and ways to fail the higher you go in levels.
- **Game Manager:**
  - The **Game Manager should be a Singleton**.
  - It should **retain the score** across levels.
  - The **Game Manager must NOT be destroyed when loading new scenes**.

AI use is NOT PERMITTED. And will be penalized if suspected.

How to submit your project :

Github link : https://github.com/revolloverr/420_541_Assignement_2026

1. Submit your changes on Github.
2. Add the link to your own fork to this document
3. Save it to your assets folder.
4. Zip the Assets folder of your Assignement with this document added in the folder
5. Submit the Zipped Assets folder trought LEA.

GRADING CRITERIA

FOR EVERY CATEGORY

Category	Points	Details
Menu System	10	Play, Quit, Pause Menu, End Screen (score + restart). Code quality
Player Mechanics	20	Movement, Jump, Double Jump (with power-up), Animations using Blend tree. Code quality
HUD & UI	10	Score display, health system, timer functionality. Code quality
Game Progression	20	At least **3 levels**, **physical end goal objects**, proper scene transitions. Code quality
Pickups & Power-ups	15	**Speed, Jump, and Score pickups**, hover & rotate, disappear & reappear. Code quality
Traps & Death System	10	Red traps remove health, falling restarts level, Game Manager retains score. Code quality
Code Structure	10	Uses **separate scripts**, Singleton for Game Manager, AnimationController used correctly. Code quality
Polish & Optimization	5	Smooth transitions, clear UI, minimal bugs.

TOTAL	**100**	-